Lec28

12 March 2026 09:20 AM

Searching

Linear

Binary

A B C D E F

Discovery

100

→ [5, 2, 1, 4, 3] →

0 1 2 3 4

target = 4

999

A D C B E

P

[2, 3, 4, 5, 6, 7, 8, 9, 10]

0 1 2 3 4 5 6 7 8

Lo mid hi

7

Lo = 0

hi = 8

mid = 4

```
while (Lo <= hi) {
    mid = (Lo + hi) / 2
    if (arr[mid] == t) {
        return mid
    }
    else if (arr[mid] > target) {
        high = mid-1
    }
    else {
        Lo = mid+1
    }
}
```

arr = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

0 1 2 3 4 5 6 7 8 9

target = 12

10 m L m L H L

```
int low = 0;
    int high = arr.length-1;
    while (low <= high) {
```

0 + 9 = 4

```
int low = 0;
    int high = arr.length-1;
    while (low<= high) {
        int mid = (low+high)/2;
        if(arr[mid] == target){
            return mid;
        }
        else if(arr[mid]> target){
            high = mid-1;
        }else{
            low = mid+1;
        }
    }
```

$\frac{0+9}{2} = 4$

$\frac{5+9}{2} = 7$

$\frac{8+9}{2} = 8$

$\frac{9+9}{2} = 9$

$x^k <= n$

$x = 1\ 2\ 3\ 4\ 5\ 6$

while $(x^k <= n)$ {

x ++

}

$x - 1$

$k = 3$

$n = 149$

$1^3 <= 149$

$2^3 <= 149$

$3^3 <= 149$

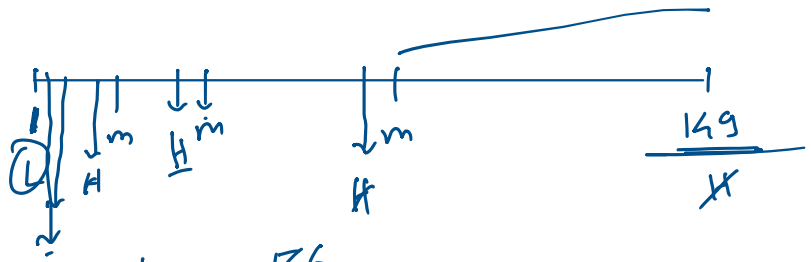
$4^3 <= 149$

$5^3 <= 149$

$6^3 <= 149$

216

$x = 0^3$



$5^3 <= 149$

Low = 6

High = 4

ans = 5

75^3

37^3

18^3

$\frac{1+149}{2} = 75$

$\frac{1+74}{2} = 37$

$\frac{1+36}{2} = 18$

$\frac{1+17}{2} = 9$

$\frac{6+6}{2} = 6$

$9^3 \le 149$

$4^3 \le 149$

$7^3 \le 149$

$5^3 \le 149$

$6^3 \le 149$

$\frac{1+9}{2} = 5$

$\frac{5+9}{2} = 7$

$\frac{5+6}{2} = 5$

$n = 149$

$L = 1$ $\quad k = 3$

$H = 149$ $\quad x =$

```
while (L <= H){
    mid = (L+H)/2
    if(mid^k <= n){
        ans = mid
        L = mid+1
    }else{
        H = mid-1
    }
}
```